FoodBar Computing & Network Infrastructure

Group 6

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FoodBar

foodbar.com

Our Company's Mission Statement

FoodBar delivers quality grocery products to its customers while maintaining secure and efficient technology systems to protect business operations and internal data.

Business Context & Security Requirements

FoodBar is a grocery retailer handling customer transactions

Critical Assets

User credentials, internal systems, database, logs

Security Priorities

Confidentiality, integrity, availability of internal services

Regulatory Influence

Payment environments typically align with PCI DSS principles

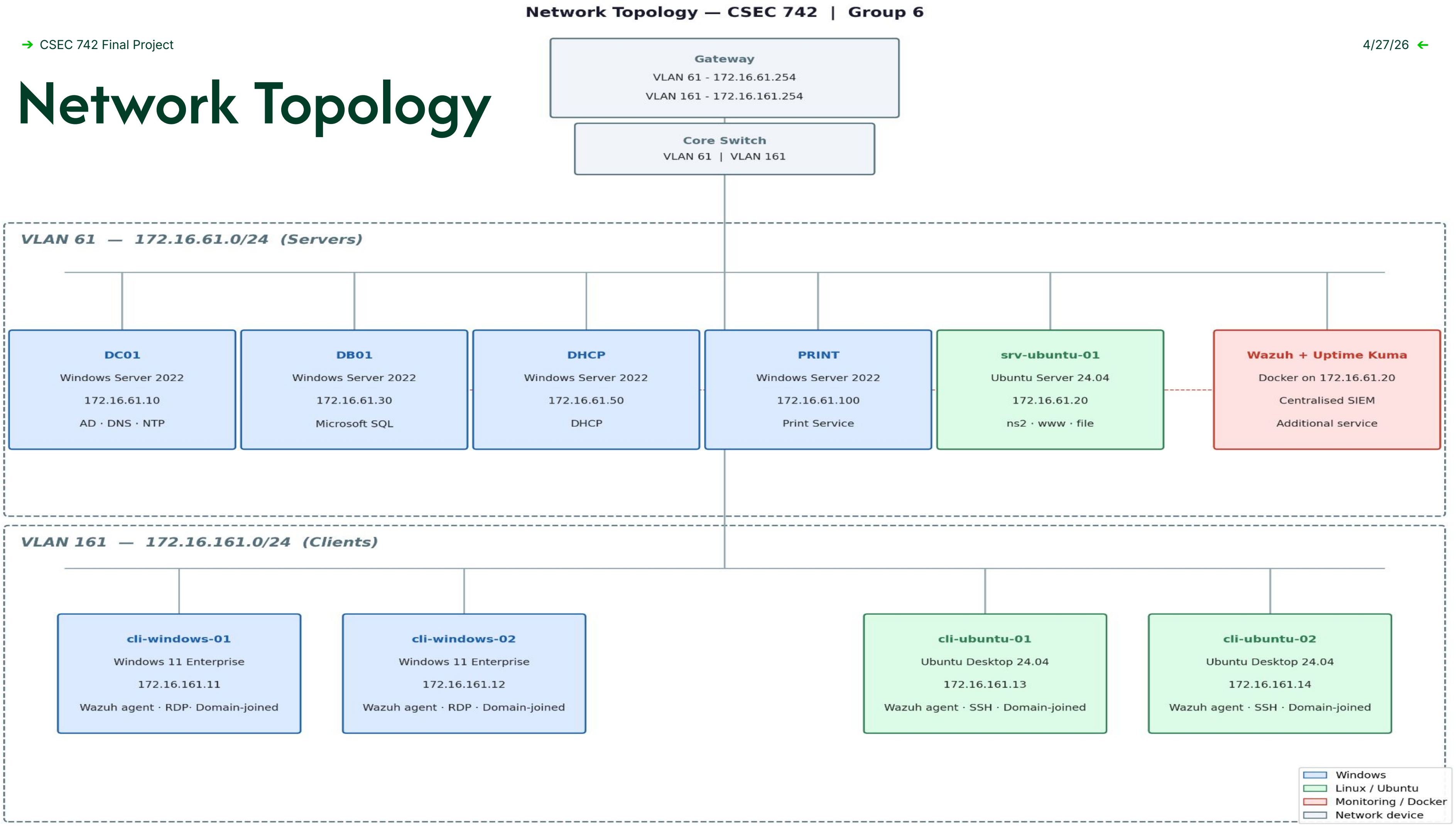
Design Impact

VLAN segmentation, AD-based authentication, centralized monitoring, restricted admin access

CSEC 742 Final Project

4/27/26

Network Topology



VM Infrastructure Overview

Servers (VLAN 61)

dc01 / ns1 / time 172.16.61.10	Active Directory Domain Controller, Primary DNS, NTP
srv-ubuntu-01 / ns2 / www / file 172.16.61.20	Secondary DNS, Web, File, Docker Container (MDR)
db01 172.16.61.30	Database
dhcp 172.16.61.50	DHCP
print 172.16.61.100	Print

Clients (VLAN 161)

cli-windows-01/02 172.16.161.11/12	Windows Clients (2)
cli-ubuntu-01/02 172.16.161.13/14	Linux Clients (2)

VLAN & Subnet Allocation

VLAN 61- Servers

Subnet: 172.16.61.0/24
Gateway: 172.16.61.254
Systems:

- .10 – AD, DNS, NTP
- .20 – Web, File, Docker, Secondary DNS, MDR
- .30 – Database Server
- .50 – DHCP Server
- .100 – Print Server

VLAN 161- Clients

Subnet: 172.16.161.0/24
Gateway: 172.16.161.254
Systems:

- .11 & .12 – Windows Clients
- .13 & .14 – Ubuntu Clients

Rationale

- Isolates critical servers from user devices
- Simplifies access control & troubleshooting
- Reduces broadcast traffic
- Supports performance optimization
- Helps meet PCI DSS segmentation requirements

System Inventory

System	OS & Version	Purpose	Services / Apps	Users
dc01 / ns1 / time	Windows Server 2022 Version 21H2	Primary Domain Controller, DNS, NTP	Active Directory Domain Services, DNS, NTP	All domain users, IT admins
srv-ubuntu-01 / ns2 / file / www	Ubuntu 24.04.4 LTS	Web server, file server, secondary DNS, Docker host	Nginx 1.24.0, vsftpd 3.0.5, BIND 9.18.39, Docker Engine 29.3.1, Wazuh Manager, Indexer & Dashboard 4.14.4, Uptime Kuma 1.23.17	Employees, IT admins
db01	Windows Server 2022 Version 21H2	Database server	Microsoft SQL Server 2022 (16.0.1000.6)	Backend services, IT admins
dhcp	Windows Server 2022 Version 21H2	DHCP server	Windows DHCP Server Role	All network devices, IT admins
print	Windows Server 2022 Version 21H2	Print services	Windows Print Services Role	Employees, IT admins
cli-windows-01 cli-windows-02	Windows 11 Version 25H2	Windows workstations	Microsoft Office, AD client services, OpenSSH 9.6p1	Employees, IT admins
cli-ubuntu-01 cli-ubuntu-02	Ubuntu 24.04.4 LTS	Linux workstations	Developer tools, OpenSSH 9.6p1	Developers, IT

OS & Application/Service Version Rationale

System Layer	Version Chosen	Why Selected	Security Benefit
Core Infrastructure	Windows Server 2022, SQL Server 2022	Enterprise LTS + AD integration	Centralized identity, access control, auditing
DNS Redundancy	Windows Server 2022 (AD-integrated) & BIND 9.18 (Ubuntu 24.04)	Cross-platform redundancy	Protects against single point of failure
Linux Services	Ubuntu 24.04 LTS, Nginx 1.24, vsftpd 3.0.5, Docker	Stable, lightweight, modular stack	Reduced attack surface, TLS, secure administration
Monitoring & Containers	Docker 29.3.1, Wazuh 4.14.4, Uptime Kuma 1.23.17	Active maintenance + modular deployment	SIEM logging, real-time alerts, isolation
Endpoints & Access	Windows 11, Ubuntu 24.04, OpenSSH 9.6p1, OpenVPN 2.6.19	Enterprise compatibility + secure access	Encrypted communications, policy enforcement, secure tunneling

Security Controls Implemented

Identity & Access Controls

→ AD DS

→ Account creation restricted to privileged groups

→ All creation events are logged

→ Centralized authentication enforces least privilege across domain-joined systems

Threat Detection and Response

→ Wazuh

→ Brute force detection across SSH, RDP, and SQL with automated alerting

→ Real time alerts on new account creation and scheduled task modifications

→ File Integrity Monitoring on web directories with CVE vulnerability scanning

Policy & Access Hardening

→ CIS Benchmarks

→ Account lockout policies and password complexity enforced domain-wide via GPO

→ Cron and scheduled task access restricted to trusted users

→ Firewall rules and service restrictions limit exposure of remote services

Secure Remote Access

→ OpenVPN

→ Eliminates direct external exposure of RDP, SSH, and SMB by tunneling remote access through VPN

→ Ensures remote connections are authenticated and encrypted

→ Reduces attack surface by preventing unauthenticated access from outside the network

Deep Dive: Wazuh Integration

Description

- Open-source SIEM and XDR platform
- Deployed as a Docker container on the Ubuntu server, providing monitoring for the entire infrastructure
- Consists of three core components: Manager, Indexer, and Dashboard
- Agents installed on all endpoints (clients and servers) which report back to the Manager

Infrastructure Integration

- Agent-based monitoring on all Windows and Linux hosts via lightweight agents
- Monitors Active Directory events
- Provides File Integrity Monitoring (FIM)
- Log collection aggregates syslog, Windows Events, and nginx logs into single dashboard
- Vulnerability Detection module scans agents against NVD CVE database

Supported CSF Categories

- **ID.AM:** Asset inventory provides real-time asset visibility
- **ID.RA:** Vulnerability Detection module maps CVEs to installed software
- **DE.AE:** Correlates events across all agents to surface anomalous patterns
- **DE.CM:** Real-time log ingestion and rule-based alerting across all hosts
- **RS.AN:** Dashboards provide alert triage, event timelines, and agent investigation

Effectiveness Evaluation

Strengths

- Covers major MITRE ATT&CK detection categories
- Built-in rules detect critical Windows Events
- CIS Benchmark module automatically scores each host against hardening standards and shows gaps

Limitations

- No network-level visibility
- Wazuh Active Response is reactive, not preventative (responds after threshold of events)

Project Challenges & Solutions

	Problem Description	Solution
Air-Gap Environment	Physical lab servers had no internet access, requiring all software, installers, and ISOs to be transferred from an internet connected machine. Missing required software disrupted progress.	Utilized Proxmox USB passthrough and scp to distribute transferred files to VMs. Retrieved needed software from identical internet-connected VMs.
Linux Package Dependencies	Ubuntu package manager (apt) is designed to resolve dependencies automatically by pulling from internet repositories. This workflow breaks in an air gapped environment.	Running apt-get download on an online machine pulls the target package and installs offline with dpkg -i (repeat until all dependencies resolve)

Project Challenges & Solutions

	Problem Description	Solution
Team Coordination	Multiple team members working simultaneously on shared infrastructure risked conflicting changes to domain-wide settings like GPO, DNS, and DHCP.	Divided infrastructure ownership by service area to prevent overlapping changes. Established check-ins and progress updates to ensure everyone stayed up-to-date on recent changes.

Thank You

Questions?